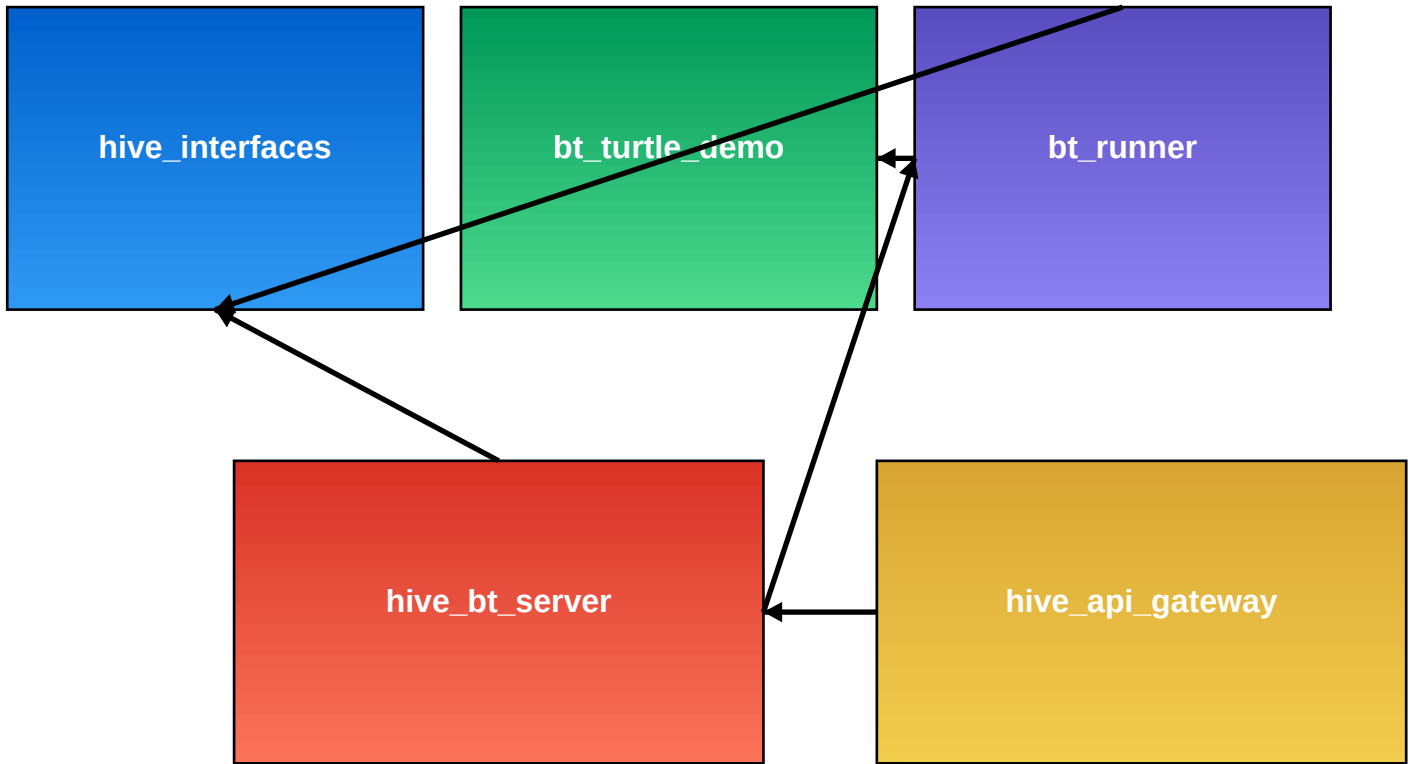


Hive BT – Code Architecture Cookbook

Packages • Files • CMake • Actions • Topics • Behavior Mapping

Package Dependency Graph



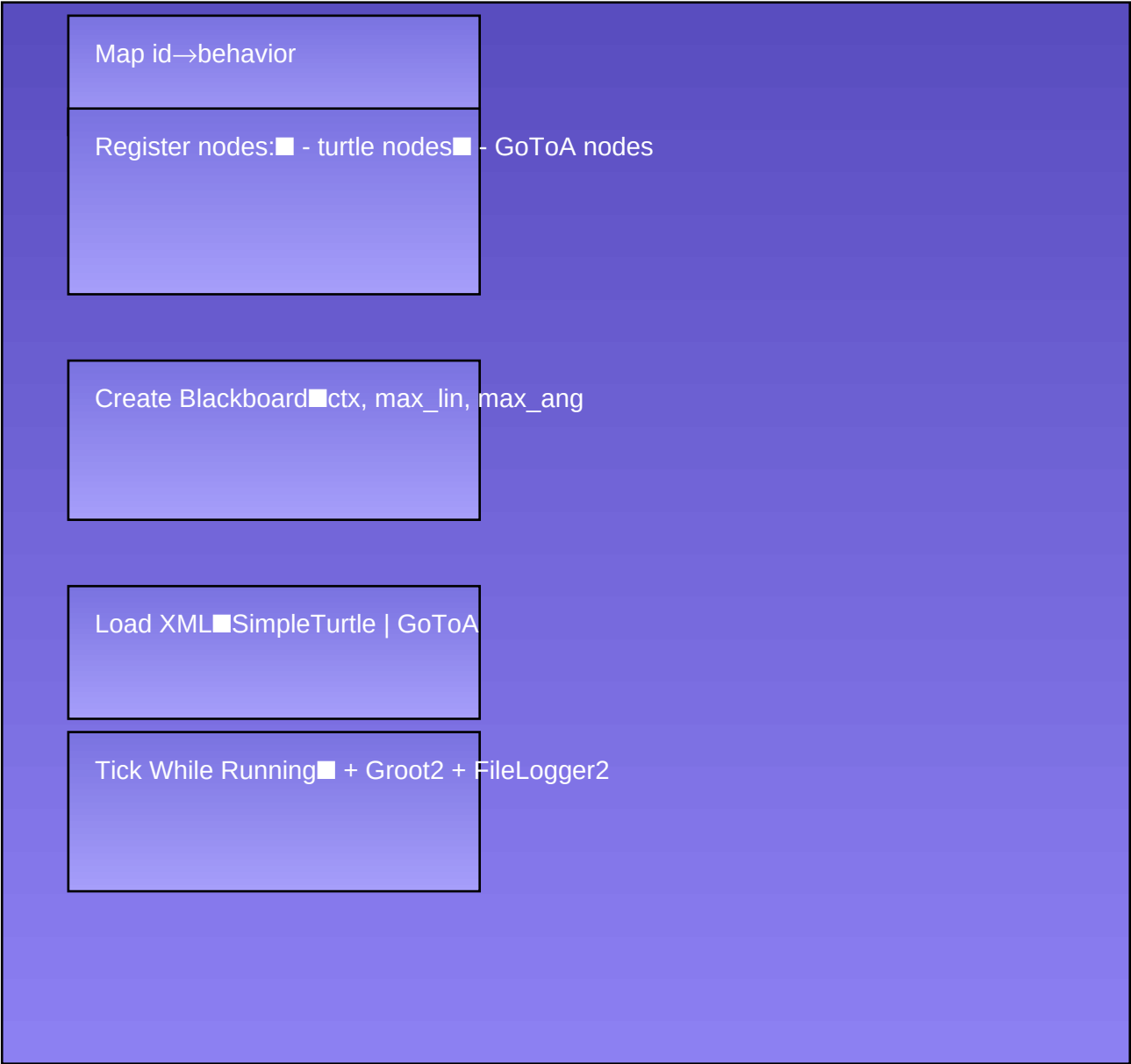
File & Node Layout

bt_runner/■ include/bt_runner/nodes_goto.hpp■

hive_bt_server/■ src/server.cpp■

bt_turtle_demo/■ include/bt_turtle_

BT-Runner Internals



Hive-BT-Server Internals

Action Server (/hive/execute_behavior)

Route id→behavior (params optional)

Async send goal to /bt_runner/execute_behavior

Feedback callback → publish /hive/status

Poll result (no extra executor) → succeed/abort/cancel

API Gateway Internals

FastAPI app + /tasks (POST)

Generate task_id + return 202

Create ExecuteBehavior.Goal

ROS 2 bridge (rclpy) → /hive/execute_behavior

Add a New Behavior (Developer Checklist)

1) Create leaves in `bt_runner/include/bt_runner/<your_nodes>.hpp`; register with BehaviorTreeFactory.

2) Write behavior XML in `bt_runner/trees/<your_tree>.xml` (compose nodes/subtrees).

3) Map id→behavior in `bt_runner/src/runner.cpp` (or accept behavior_name from API).

4) CMake: install `include/` + `trees/` and `ament_export_include_directories(include)`.

5) Build with colcon; source `install/setup.bash`.

6) Test: POST `/tasks {"id":<new_id>}` and watch Groot2.

Common Issues & Fixes

Package not found:

Installament index marker + package.xml; re-source install/setup.bash.

• Header not found:

Export include dirs & target_include_directories; install include/.

• Node added to executor:

Avoid spin_until_future_complete on a spinning node; use async + wait_for.

• BT port type mismatch:

Use consistent port types (e.g., int64_t).

• No /amcl_pose:

Switch WaitAtTarget to use /tf odom distance or Nav2 action feedback.